

Muaaz Bhamjee

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PROFILE

Associate Professor and Founder / Principal Investigator of the CERG-FLUX Lab (Fluids, Learning, and Uncertainty in complex systems) at the University of Pretoria, with a background in Mechanical Engineering and Applied Mathematics. Research spans computational fluid dynamics and multiphase / granular flow (Navier-Stokes and Lattice Boltzmann methods on high-performance computing), scientific machine learning and AI for climate, experimental high-energy physics with the ATLAS Collaboration at CERN, and hybrid quantum computing and machine learning.

EDUCATION

Doctor of Engineering (DIng, Mechanical Engineering Science) 2011–2017
University of Johannesburg

Thesis: Modelling of the Multiphase Interactions in a Hydrocyclone using Navier-Stokes and Lattice Boltzmann based Computational Approaches.

Master of Engineering (MIng, Mechanical Engineering Science), *Cum Laude* 2009–2011
University of Johannesburg

Dissertation: A Computational Fluid Dynamics and Experimental Investigation of an Airflow Window.

BSc Honours (Applied Mathematics), *Cum Laude* 2008–2010
University of Johannesburg

Bachelor of Engineering (BIng, Mechanical Engineering Science) 2004–2008
University of Johannesburg

ACADEMIC AND PROFESSIONAL POSITIONS

Associate Professor, *University of Pretoria* Feb 2023 – Present
Founder and Principal Investigator, CERG-FLUX Lab. Teaching (MIA 320), postgraduate supervision, and research in computational mechanics, scientific machine learning, and quantum technologies.

Staff Research Scientist, *IBM Research – Africa* Feb 2023 – Dec 2024
Data-driven and machine-learning approaches to climate; challenge / project lead; patent ideation; student and intern mentorship.

Senior Research Associate (Visiting), *University of Johannesburg* Jul 2023 – Dec 2024
Research and postgraduate supervision.

Senior Lecturer, *University of Johannesburg* May 2017 – Jan 2023
Course co-ordination and lecturing; postgraduate supervision; CFD and engineering-education research; former co-ordinator of the MEng (Sustainable Energy) programme.

Lecturer, *University of Johannesburg* Jan 2015 – Apr 2017
Course co-ordination and lecturing; final-year project supervision; co-supervision of Masters students.

Junior Engineer (CFD Analyst), *Hatch Africa (Pty) Ltd* 2008 – 2009
CFD modelling for the metals and mineral-processing environment; user-defined functions; technical reporting.

RESEARCH LEADERSHIP AND PROFESSIONAL AFFILIATIONS

- Founder and Principal Investigator, CERG-FLUX Lab, University of Pretoria (2026–present).
- Principal Investigator, SA-CERN ATLAS Project; University of Pretoria ATLAS / CERN Institutional Representative and Team Leader.

- Registered Professional Engineer (PrEng), Engineering Council of South Africa (ECSA).
- Vice-President, SAAM Executive Committee; Vice-President, South African National IUTAM Committee.
- Member, South African Institute of Mechanical Engineering (SAIMechE) and the South African Association for Theoretical and Applied Mechanics (SAAM).

NOTEWORTHY RESEARCH GRANTS AND FUNDING

- **2026 – UPQuST / SA QuTI-linked grants:** Hybrid Quantum-Classical Accelerated CFD (R180,000); Quantum Machine Learning for Rare Particle Searches in HEP (R180,000); Quantum-Magnetic Sensors for Particle Detection in Mineral Separation (R400,000).
- **2022 – UJ URC Research Equipment Grant:** Preventing COVID-19 with Mathematical Modelling, Digital Twins, AI and 4IR Engineered Solutions (R284,517).
- **2021 – UJ GES COVID-19 Grant:** same programme (R80,000).
- **2020–2021 – European Synchrotron Radiation Facility (ESRF):** Equipment for the BEATS beamline at SESAME and ESRF beamlines (€8,750).
- **2017–2019 – NRF Thuthuka Funding Instrument (Post-PhD Track):** Modelling of Multiphase Flow in Process Equipment (TTK160615171464; R602,000).
- **Patent:** Dynamically Forecasting High-Resolution Air Temperature in Real-Time Using Multiple Sources – ZA2023/09448 (granted 30 July 2024); USA pending (US18/200,050).
- **CHPC high-performance computing allocations:** Multiphase Flow Modelling and Deep Learning (MECH1739, active); Modelling of Multiphase Flow in Process Equipment (MECH0837).

RESEARCH OUTPUT AND POSTGRADUATE SUPERVISION

Publication record indexed on Google Scholar (h-index 65), Scopus (h-index 34) and Web of Science (h-index 30), spanning journal articles, ATLAS Collaboration papers, conference papers, a book chapter and a granted patent. Full publication list available separately. Postgraduate supervision: 9 Masters graduated; currently main or co-supervisor of 8 doctoral and 3 Masters candidates.

CURRENT DOCTORAL SUPERVISION

- M. Chambakata, “Lattice Boltzmann Method for Modelling Two-Phase Flow” (main supervisor).
- A.Y. Akinbowale, “Quantum Machine Learning for Rare Particle Signatures in High-Dimensional Physics Data” (main supervisor).
- M. Ramokali, “Hybrid Classical-Quantum Computing and Machine Learning Methods for Computational Fluid Dynamics” (main supervisor).
- M.T. Saphira Agnes, “Fluid-Structure Interactions in Pathological Arteries: Numerical and Experimental Models” (main supervisor).
- M.S.W. Potgieter, “Hybrid Lattice Boltzmann – Machine Learning Methods for Two-Phase Flow” (main supervisor).
- A. Essop, “A Novel Numerical Scheme for High-Speed Flows” (main supervisor).
- T. Jooma Abbajee, “Graph Dynamical Reservoir Computing” (co-supervisor).
- M.P. Connell, “Search for Higgs Boson Decays via Dark Bosons to a Four-Lepton Final State” (co-supervisor).

SELECTED POSTGRADUATE SUPERVISION (GRADUATED)

- T. Jooma Abbajee, “Reservoir Computing: A Tool for Predicting Chaotic Dynamical Systems,” University of Johannesburg, 2026 (co-supervisor).
- K.M. Makhanya, “The Use of CFD for Assessing Flow-Induced Acoustics to Diagnose Lung Conditions,” University of Johannesburg, 2025 (co-supervisor).
- T.F. Mokoena, “Experimental Hutch Equipment for the BEATS Beamline at SESAME,” University of Johannesburg, 2022 (main supervisor).
- T. Ramokoka, “Fluid and Particulate Flow Through a Ventriculoperitoneal Shunt in a Variable-Temperature Environment,” University of Johannesburg, 2022 (sole supervisor).
- A. Potgieter, “Modelling of a Heated Gas-Solid Fluidised Bed using Eulerian Based Models,” University of Johannesburg, 2020, *Cum Laude* (main supervisor).

SELECTED PUBLICATIONS

- ATLAS Collaboration, “Observation of quantum entanglement with top quarks at the ATLAS detector,” *Nature* **633**, 542–547 (2024).
- ATLAS Collaboration, “Characterising the Higgs boson with ATLAS data from the LHC Run-2,” *Physics Reports* **1116**, 4–56 (2025).
- “Infectiousness model of expelled droplets exposed to ultraviolet germicidal irradiation,” *Computers & Fluids* **275**, 106242 (2024).
- “Experimental and CFD Investigation of a Hybrid Solar Air Heater,” *Solar Energy* **195**, 413–428 (2020).
- “An Experimentally Validated Mathematical and CFD Model of a Supply Air Window: Forced and Natural Flow,” *Energy and Buildings* **57**, 289–301 (2013).
- “Detection and Characterization of Urban Heat Islands with Machine Learning,” *IGARSS 2024 – IEEE International Geoscience and Remote Sensing Symposium*, Athens, 1693–1699 (2024).
- “Reservoir Computing for Predicting Chaotic Dynamical Systems,” *69th Annual Conference of the South African Institute of Physics (SAIP 2025)*, 1026–1031 (2025).
- “The Modification of the Dynamic Behaviour of Cyclonic Flow in a Hydrocyclone under Surging Conditions,” *Mathematical and Computational Applications* **27**(6), 88 (2022).

SELECTED PROFESSIONAL SERVICE

- Journal reviewing: *Physics of Fluids* (AIP), *Applied Energy* (Elsevier), *Energy and Buildings* (Elsevier), *Environmental Science and Pollution Research* (Springer), and the *SAIMechE R&D Journal*; conference reviewing for SAIP and SACAM, and ICLR 2024 workshop proposals.
- Conference organising committees: 7th Biennial Workshop on Discovery Physics at the LHC (Kruger2026, SA-CERN); 2025 SAIMechE Central Branch Conference (Wits); SMPM 2019; SACAM 2012.
- Postgraduate examination: external examiner for doctoral and Masters candidates at the University of Cape Town, the University of Johannesburg and the University of KwaZulu-Natal; internal examiner at the University of Pretoria.